Krishnaprasad K.A.

Gainesville, FL krishnap561997@gmail.com LinkedIn GitHub Google Scholar

RESEARCH PROFILE

My research seeks to develop predictive computational frameworks that preserve the essential physics of complex transport phenomena while maintaining predictive accuracy at substantially reduced computational cost. By integrating interface-resolved simulations, turbulence modeling, asymptotic theory, depth-integrated modeling, uncertainty quantification, and scientific machine learning, I develop multiscale models that translate high-fidelity numerical simulations into efficient predictive tools for environmental and engineering flows.

EDUCATION

University of Florida Gainesville, FL
Doctor of Philosophy in Mechanical Engineering. Jan. 2022 - Dec. 2025
Advisor: Prof. S. Balachandar
Dissertation: *Predictive Modeling and Uncertainty Quantification of Indoor Aerosol Transport*

University of Florida Gainesville, FL
Master of Science in Mechanical Engineering. Jan. 2021 - May 2023

National Institute of Technology, Trichy Tiruchirapalli, India
Bachelor of Technology in Mechanical Engineering. Aug. 2015 - May 2019

EXPERIENCE & EMPLOYMENT HISTORY

Postdoctoral Researcher | University of Florida Jan, 26 - Present

- Lead computational modeling efforts within an ONR-funded program focused on predictive modeling of hydrodynamic instabilities and heat transfer in microchannel flow boiling.
- Developed interface-resolved direct numerical simulations to elucidate the physics of hydrodynamic instabilities and provide high-fidelity datasets for model development and validation.
- Formulated physics-based reduced-order models that retain the predictive capability of interface-resolved simulations while reducing computational cost by more than two orders of magnitude.
- Developed hybrid machine-learning closure frameworks coupled with physics-based reduced-order solvers to extend predictive accuracy beyond the asymptotic regimes of validity of theoretical closures.

Graduate Research Assistant | University of Florida Aug, 21 - Dec, 25

- Developed predictive computational frameworks for turbulent aerosol transport in mechanically ventilated indoor environments using massively parallel Euler–Lagrange Large-Eddy Simulations, enabling physics-based assessment of airborne contaminant transport across realistic ventilation configurations.
- Pioneered a novel statistical overloading methodology aimed at capturing turbulent dispersal, uncertainty estimation, and EL simulation design to obtain comprehensive two-point two-time Lagrangian statistics.
- Developed scalable scientific software and automated post-processing workflows capable of analyzing aerosol transport statistics across approximately 25 million source–receiver pairs generated from EL-LES data.
- Published 10 peer-reviewed journal articles spanning turbulent aerosol transport, uncertainty quantification, indoor air quality, and bedload sediment transport while collaborating with academic and industrial partners.

Student Research Assistant | University of Florida Jan, 21 - Jul, 21

- Developed 3D CFD models to study transport and waste clearance in perfusion bioreactors for hepatic tissues.
- Performed parametric studies to optimize nutrient delivery and waste removal protocols for experimental design.

Research Intern | IIT, Kharagpur Aug, 19 - May, 20

- Worked on analytical formulations to predict magnetohydrodynamic deformation of viscoelastic fluids.
- Performed validation studies of the analytical model using computational simulations in ANSYS Fluent.

RESEARCH DIRECTIONS

Predictive Simulation of Turbulent Transport Phenomena 2021 – Present

- Developed EL-LES frameworks in Nek5000 for modeling turbulent aerosol transport in indoor environments.
- Extended the computational framework to incorporate HVAC filtration, thermal plumes, natural convection, and multiple ventilation configurations, enabling physics-based assessment of airborne contaminant transport.
- Collaborated with researchers at Pusan National University to benchmark RANS models against LES across multiple ventilation configurations and coordinated experimental validation studies with LG Electronics.

Statistical Modeling and Uncertainty Quantification 2023 – Present

- Introduced the statistical overloading methodology for rigorous uncertainty quantification in Euler–Lagrange simulations, providing computationally efficient estimation of two-point, two-time dispersal statistics.
- Developed analytical and empirical models relating prediction uncertainty to particle count, sampling duration, and turbulent transport characteristics, establishing practical computational design guidelines.
- Developed scalable scientific software and automated workflows for statistical analysis of large CFD datasets spanning approximately 25 million source–receiver pairs.

Physics-Based Reduced-Order Modeling of Interfacial Flows 2024 – Present

- Develop interface-resolved direct numerical simulations to investigate hydrodynamic instabilities, interfacial transport, and heat-transfer mechanisms governing thin liquid films in annular flows.
- Formulate reduced-order models that retain the predictive capability of interface-resolved simulations while reducing computational cost by more than two orders of magnitude.
- Integrate asymptotic theory and scientific machine learning to develop robust data-driven closure models that extend predictive capability across broad operating regimes.

SKILLS

Computational Fluid Dynamics: Nek5000, Basilisk C, ANSYS Fluent, OpenFOAM, COMSOL
Programming Languages: C/C++, Fortran, Python, MATLAB, Maple
High-Performance Computing: MPI, OpenMP, CUDA, Slurm, Parallel Scientific Computing
Scientific Computing: Finite Volume Methods, Spectral Element Methods, Interface-Resolved DNS, Euler–Lagrange LES, Reduced-Order Modeling, Scientific Machine Learning
Visualization and Post-processing: ParaView, MATLAB, Python
Engineering Software: SolidWorks, AutoCAD

PUBLICATIONS

[1] I. Velazquez, **K.A. Krishnaprasad**, N. Zgheib, C. McGhee, C. Ferguson, D.C.J.D. Hoyal, S. Balachandar, "Oscillatory flow around a periodic array of circular cylinders under bedload dominated sediment transport," Physics of Fluids (2025).

[2] **K.A. Krishnaprasad**, N. Zgheib, S. Balachandar, "A statistical approach to turbulent dispersal of aerosols for accurate prediction of concentration and associated uncertainties," Physical Review Fluids (2025).

[3] **K.A. Krishnaprasad**, R. Patel, C. El Khoury, A.J. Banko, N. Zgheib, S. Balachandar, "The statistical overloading framework for accurate evaluation of pollutant dispersal with rigorous uncertainty estimation," Journal of Aerosol Science (2025).

[4] R. Patel, **K.A. Krishnaprasad**, N. Zgheib, J.S. Salinas, S. Balachandar, "Influence of internal heating and natural convection on airborne transmission of viral diseases," Fluid Dynamics Research (2024).

[5] M.R. Anjum, V. Subramaniam, B.R. Higgins, C. Abrahan, S. Chisolm, **K.A. Krishnaprasad**, O. Azie, G. Palmer, T. Angelini, M. Sarntinoranont, "Determining Rates of Molecular Secretion from Supernatant Concentration Measurements in a 3D-Bioprinted Human Liver Tissue Model," ACS Biomaterials Science and Engineering (2024).

[6] **K.A. Krishnaprasad**, N. Zgheib, M.Y. Ha, K. Choudhary, C.Y. Choi, S. Jang, K.S. Bang, S. Balachandar, "Existence of a non-zero worst-case ACH for short-term exposure in ventilated indoor spaces," Indoor Air (2024).

[7] K. Choudhary, **K.A. Krishnaprasad**, N. Zgheib, M.Y. Ha, S. Balachandar, "Effect of room size, shape, AC placement, and air leakage on indoor airborne viral transmission," Building and Environment (2023).

[8] K. Choudhary, **K.A. Krishnaprasad**, S. Pandey, N. Zgheib, J.S. Salinas, M.Y. Ha, S. Balachandar, "Effectiveness of RANS in predicting indoor airborne viral transmission: A critical evaluation against LES", Computers & Fluids (2023).

[9] **K.A. Krishnaprasad**, J.S. Salinas, N. Zgheib, S. Balachandar, "Fluid mechanics of air recycling and filtration for indoor airborne transmission," Physics of Fluids (2023).

[10] J.S. Salinas, **K.A. Krishnaprasad**, N. Zgheib, S. Balachandar, "Improved guidelines of indoor airborne transmission taking into account departure from the well-mixed assumption," Physical Review Fluids (2022).

CONFERENCE PRESENTATIONS

APS Division of Fluid Dynamics (APS DFD) Meeting

- **K.A. Krishnaprasad**, R. Patel, A.J. Banko, N. Zgheib, and S. Balachandar. "The statistical overloading framework for dispersal problems: A comprehensive evaluation of the computational requirements." Houston, TX, USA, 2025.
- R. Patel, **K.A. Krishnaprasad**, N. Zgheib, and S. Balachandar. "Large Eddy Simulation-Based Optimization of Occupant Configurations in Pre-Existing Indoor Spaces for Mitigating Airborne Pathogen Transmission." Houston, TX, USA, 2025.
- **K.A. Krishnaprasad**, R. Patel, K. Choudhary, N. Zgheib, J. Salinas, M.Y. Ha, and S. Balachandar. "Predicting aerosol dispersal in indoor spaces and the associated uncertainties due to turbulence." Salt Lake City, UT, USA, 2024.
- R. Patel, **K.A. Krishnaprasad**, N. Zgheib, J. Salinas, and S. Balachandar. "Fluid Mechanics-Informed Statistical Optimization of Spatial Configurations to Minimize Airborne Disease Transmission in Indoor Spaces." Salt Lake City, UT, USA, 2024.
- **K.A. Krishnaprasad**, N. Zgheib, J. Salinas, K. Choudhary, M.Y. Ha, and S. Balachandar. "Risk assessment using a fluid-mechanics informed statistical framework for short- and long-term exposure for indoor airborne viral transmission." Washington, DC, USA, 2023.
- R. Patel, **K.A. Krishnaprasad**, N. Zgheib, J. Salinas, and S. Balachandar. "A statistical framework using LES to assess the effect of internal heating and natural convection on airborne transmission." Washington, DC, USA, 2023.
- **K.A. Krishnaprasad**, N. Zgheib, J. Salinas, and S. Balachandar. "A statistical framework for assessing the effectiveness of filtration and ventilation in preventing indoor airborne viral transmission using high-fidelity simulations." Indianapolis, IN, USA, 2022.

Invited Talks

- **K.A. Krishnaprasad**, N. Zgheib, J. Salinas, and S. Balachandar. "Exploring the effects of filtration, room geometry, and heating in indoor airborne viral contagion using high-fidelity simulations." *Poster Presentation*, Gordon Research Conference on Fluids in Disease Transmission and Contamination, South Hadley, MA, USA, 2022.

Other Conferences

- S. Balachandar, **K.A. Krishnaprasad**, R. Patel, K. Choudhary, N. Zgheib, J. Salinas, M.Y. Ha. "Towards a rapid prediction framework for aerosol transmission of diseases in indoor spaces." International Congress of Theoretical and Applied Mechanics (ICTAM), Daegu, South Korea, 2024.
- **K.A. Krishnaprasad**, N. Zgheib, J. Salinas, and S. Balachandar. "A statistical approach to viral airborne risk assessment using high-fidelity simulations." ASTFE Thermal and Fluids Engineering Conference, Las Vegas, NV, USA, 2022.
- K. Choudhary, **K.A. Krishnaprasad**, S. Pandey, S. Balachandar, and M.Y. Ha. "Flow characteristics of a four-way air-conditioner cassette within a room containing a window." Korean Society of Mechanical Engineers Spring and Autumn Conference, Jeju, South Korea, 2022.

ACHIEVEMENTS

Renwick Scholarship 2020
Recipient of the Glenn and Deborah Renwick Engineering Scholarship awarded to outstanding master's students who demonstrate a strong background in academics and research

National Talent Search Scholarship 2013
Recipient of the National Talent Search Scholarship awarded to outstanding students throughout the country after 2 rounds of examinations

MENTORING

Sami Latouf, Undergraduate Student, University of Florida 2026
Javier Camacho, Undergraduate Student, University of Florida 2026
Irvin Velazquez, Graduate Student, University of Texas, Rio-Grande Valley 2024 - 2025
Rupal Patel, Graduate Student, University of Florida 2022 - 2026
Kailash Choudhary, Graduate Student, Pusan National University 2022 - 2024
Charbel El Khoury, Undergraduate Student, Lebanese American Universy 2022

TEACHING EXPERIENCE

Teaching Assistant, Fluid Mechanics - 1 Fall, 2026